

much less marked differences in employment growth across AI exposure quintiles.¹⁵

4.2 Fact 2: Though overall employment continues to grow, employment growth for young workers has been stagnant

Overall employment in the ADP data remains robust, coinciding with a low national unemployment rate in the post-pandemic period. However, Figure A4 suggests some leveling off in employment growth for young workers relative to other age groups, consistent with recent discussion of a worsening job market for entry-level workers (Chen, 2025; Federal Reserve Bank of New York, 2025).

Figure A5 offers insight into how these trends relate to AI exposure. For each age group, employment growth from late 2022 to September 2025 was 5-13% for the lowest three AI exposure quintiles, with no clear ordering in employment growth by age. In contrast, for the highest two exposure quintiles employment for 22-25 year olds declined by 6% between late 2022 and September 2025, while employment for workers aged 35-49 grew by over 8%. These results show that declining employment in AI-exposed jobs is driving tepid overall employment growth for workers between the ages of 22 and 25.

While these findings suggest divergent employment outcomes by AI exposure for young workers, we caution the trends observed in these first two facts could be driven by other changes in the US economy. Our subsequent facts evaluate the robustness of the results to alternative analyses.

4.3 Fact 3: Entry-level employment has declined in applications of AI that *automate* work, with muted changes for *augmentation*

AI exposure can either complement or substitute for labor. These may have very different implications for the labor market (Brynjolfsson, 2022).

To assess how employment patterns differ based on the complementarity or substitutability of AI with labor, we use data on generative AI usage from the Anthropic Economic Index (Handa et al., 2025). The Index provides an estimate of the share of queries that pertain to each occupation. In addition, for each task it reports estimates of the share of queries pertaining to that task that are “automative,” “augmentative,” or none of the above. We use these classifications as estimates

¹⁵Figure A3 shows that declining employment for young workers spans a range of occupations. Close to 70% of occupations in the first exposure quintile see rising early career employment between October 2022 and September 2025, compared to less than half of occupations in the fifth quintile.

occupation cells. Future work should study employment trends in other large-scale data sources such as the American Community Survey (ACS) or data from Revelio Labs as in [Hosseini and Lichtinger \(2025\)](#) and [Klein Teeselink \(2025\)](#).

5 Conclusion

We document six facts about the recent labor market effects of artificial intelligence.

- We find substantial employment declines for early-career workers in occupations most exposed to AI, such as software development and customer support.
- While economy-wide employment continues to grow, employment growth for young workers has been stagnant.
- Entry-level employment has declined in applications of AI that *automate* work, with muted effects for those that *augment* it.
- After conditioning on firm-time effects, young workers experienced a 16% relative employment decline in the most exposed occupations.
- Labor market adjustments are more visible in employment than in compensation.
- These patterns hold across various alternative analyses.

While our main estimates may be influenced by factors other than generative AI, our results are consistent with the hypothesis that generative AI has begun to affect entry-level employment significantly.

The adoption of new technologies typically leads to heterogeneous effects across workers, resulting in adjustment periods as workers reallocate from displaced forms of work to new forms with growing labor demand ([Autor et al., 2024](#)). Past transitions such as the IT revolution ultimately led to robust growth in employment and real wages following physical and human capital adjustments, with some workers benefiting more than others ([Bresnahan et al., 2002](#); [Brynjolfsson et al., 2021](#)).

Tracking employment trends on an ongoing basis will help determine if the adjustment to AI follows a similar pattern. We will continue monitoring these outcomes to assess whether the trends